THE MARRIAGE OF TWO SERIES: AN EXCITING APPROACH TO OBTAINING DEFINITE INTEGRAL SOLUTIONS



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In this presentation we will establish an unusual methodology for evaluating non-trivial definite integrals, some of which may or may not be new. It is unusual because we do not begin with an integral and then evaluate the integral. Instead, we go through a process of analysis to obtain the answer first and then obtain the integral that it corresponds to.

FOURIER SERIES DEFINITION

$$f(x) = \underbrace{a_0 + \sum_{n=1}^{\infty} a_n cos(nx)}_{cosine\ series} + \underbrace{\sum_{n=1}^{\infty} b_n sin(nx)}_{sine\ series}$$

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx \qquad a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx$$

TAYLOR SERIES DEFINITION

Recall the definition of the Taylor Series expansion of a function centered at $x = a$ [Stewart]:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

METHODOLOGY: AN EXAMPLE

We will examine the function $f(z) = e^z$ in this example, and use three steps for our methodology.

$$e^{\left(re^{i\theta}\right)} = e^{r\cos\theta + ir\sin\theta} = e^{r\cos\theta}e^{ir\sin\theta}$$
$$= e^{r\cos\theta}\cos\left(r\sin\theta\right) + ie^{r\cos\theta}\sin\left(r\sin\theta\right)$$

1 Decompose the function into real and imaginary parts by setting $z = re^{i\theta}$

$$e^z = 1 + \sum_{n=1}^{\infty}\frac{r^n}{n!}\cos n\theta + i\sum_{n=1}^{\infty}\frac{r^n}{n!}\sin n\theta$$

2 Take the Taylor series of the function and set $z = re^{i\theta}$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n\cos nx + i\sum_{n=1}^{\infty} b_n\sin nx$$

3 Take the above two lines and equate to the Definition of the Fourier Series. The coefficients are the closed form expressions for the coefficient integrals of the Fourier Series.

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx \Rightarrow 1 = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{r\cos\theta}\cos(r\sin\theta)\,d\theta$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx \Rightarrow \frac{r^n}{n!} = \frac{1}{\pi}\int_{-\pi}^{\pi} e^{r\cos\theta}\cos(r\sin\theta)\cos(n\theta)\,d\theta$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx \Rightarrow \frac{r^n}{n!} = \frac{1}{\pi}\int_{-\pi}^{\pi} e^{r\cos\theta}\sin(r\sin\theta)\sin(n\theta)\,d\theta$$

We equate the coefficients in (2) with the coefficients in the Fourier Series definition to obtain three very complicated integrals.